```
Kenya, step 2 (2026-11-02)

 chain     seed iterations.saved acceptance.rate final.log.likelihood best.log.likelihood errored.proposals
     1 20263016             5000           0.277             -39791.4            -39775.4                 0
     2 20263017             5000           0.277             -39827.9            -39816.0                 0
     3 20263018             5000           0.276             -39770.1            -39767.5                 0
     4 20263019             5000           0.276             -39723.0            -39712.5                 0
                                  file
 mcmc_kenya_step2_chain1_2026-11-01.Rdata
 mcmc_kenya_step2_chain2_2026-11-01.Rdata
 mcmc_kenya_step2_chain3_2026-11-01.Rdata
 mcmc_kenya_step2_chain4_2026-11-01.Rdata

Worst R-hats (> 1.1 suggests the chains haven't converged):
   age.15.to.19.transmission.multiplier.3    age.5.to.14.hiv.mortality.multiplier.0                 log.OR.suppression.slope
                                    2.752                                    2.675                                    2.305
                                  trate.3    age.30.to.39.transmission.multiplier.3    age.20.to.29.transmission.multiplier.1
                                    2.304                                    2.202                                    2.145
       over.50.hiv.mortality.multiplier.2    age.30.to.39.transmission.multiplier.1                age.25.to.50.aging.factor
                                    2.084                                    1.988                                    1.962
   age.20.to.29.transmission.multiplier.2                                   trate.4   age.15.to.24.hiv.mortality.multiplier.1
                                    1.937                                    1.930                                    1.921
               female.to.male.multiplier    age.15.to.19.transmission.multiplier.2   age.15.to.24.hiv.mortality.multiplier.0
                                    1.903                                    1.891                                    1.822
                                  trate.2                                  trate.1    age.15.to.19.transmission.multiplier.0
                                    1.813                                    1.789                                    1.694
age.50.and.over.transmission.multiplier.3    age.0.to.14.hiv.mortality.multiplier.2    age.30.to.39.transmission.multiplier.0
                                    1.683                                    1.676                                    1.662
   age.15.to.19.transmission.multiplier.1    age.20.to.29.transmission.multiplier.0    age.30.to.39.transmission.multiplier.2
                                    1.657                                    1.626                                    1.625
  age.25.to.49.hiv.mortality.multiplier.2
                                    1.562

Merged simset:
 location simulations total.weight                                    file
    Kenya        1000         1000 simset_kenya_step2_chains1234_2026-11-01.Rdata

Fit check (merged simset vs every calibration target, 2000-2025): 58 of 63 targets flagged (worst 30 shown; all in <location>_fit_check.csv)
     data.type         age    sex years.with.data pct.years.in.sim.95 median.sim.vs.data last.year last.sim.vs.data
 hiv.mortality       25-49 female              26                   0              x0.29      2025            x0.08
 hiv.mortality       25-49   both              26                   0              x0.46      2025            x0.07
 hiv.mortality 50 and over   both              26                  23              x2.06      2025            x1.04
 hiv.mortality 50 and over   male              26                   8              x1.94      2025            x1.05
 hiv.mortality 50 and over female              26                  27              x1.96      2025            x1.04
 hiv.mortality       25-49   male              26                   4              x0.62      2025            x0.05
 hiv.mortality       15-24   male              26                   8              x1.57      2025            x0.09
    prevalence 50 and over female              26                   0              x1.39      2025            x1.18
     awareness        0-14   both              11                   9          -15.6 pts      2025         -1.0 pts
 hiv.mortality       15-24   both              26                  19              x1.43      2025            x0.26
 hiv.mortality    All ages female              26                  27              x0.72      2025            x0.48
    prevalence 50 and over   male              26                  23              x0.77      2025            x0.81
    engagement         15+ female              11                   0           +9.3 pts      2025         +1.7 pts
     awareness         15+   male              11                   0           +8.7 pts      2025         +7.1 pts
    engagement         15+   both              11                   9           +9.2 pts      2025         +0.7 pts
     incidence       10-19   male              26                  65              x1.38      2025            x0.98
    engagement    All ages   both              11                   0           +8.9 pts      2025         +0.6 pts
```

Kenya step 2: likelihood

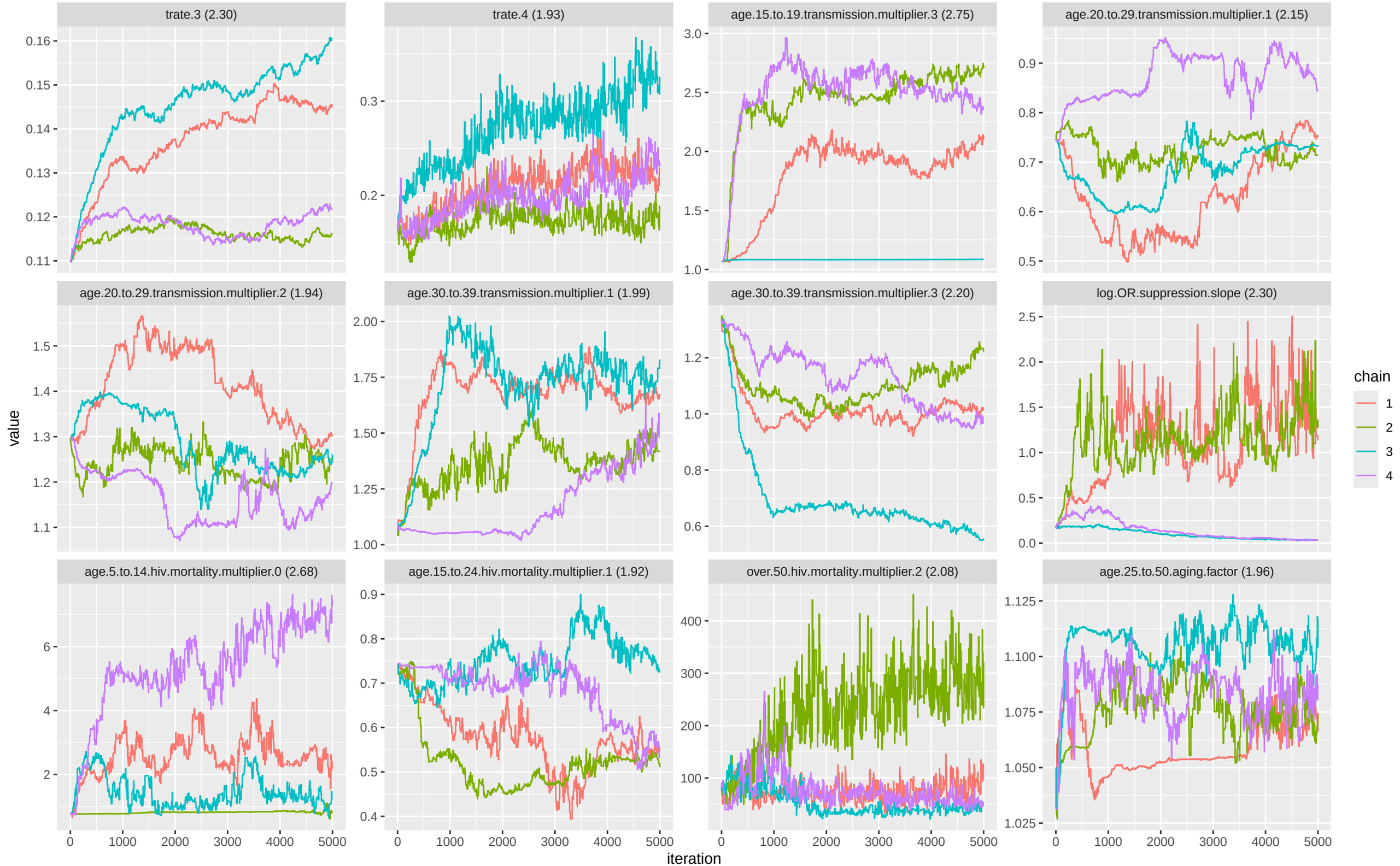
0
−10000
−20000
−30000
−40000
lue

0
25000
50000
75000
100000
iteration

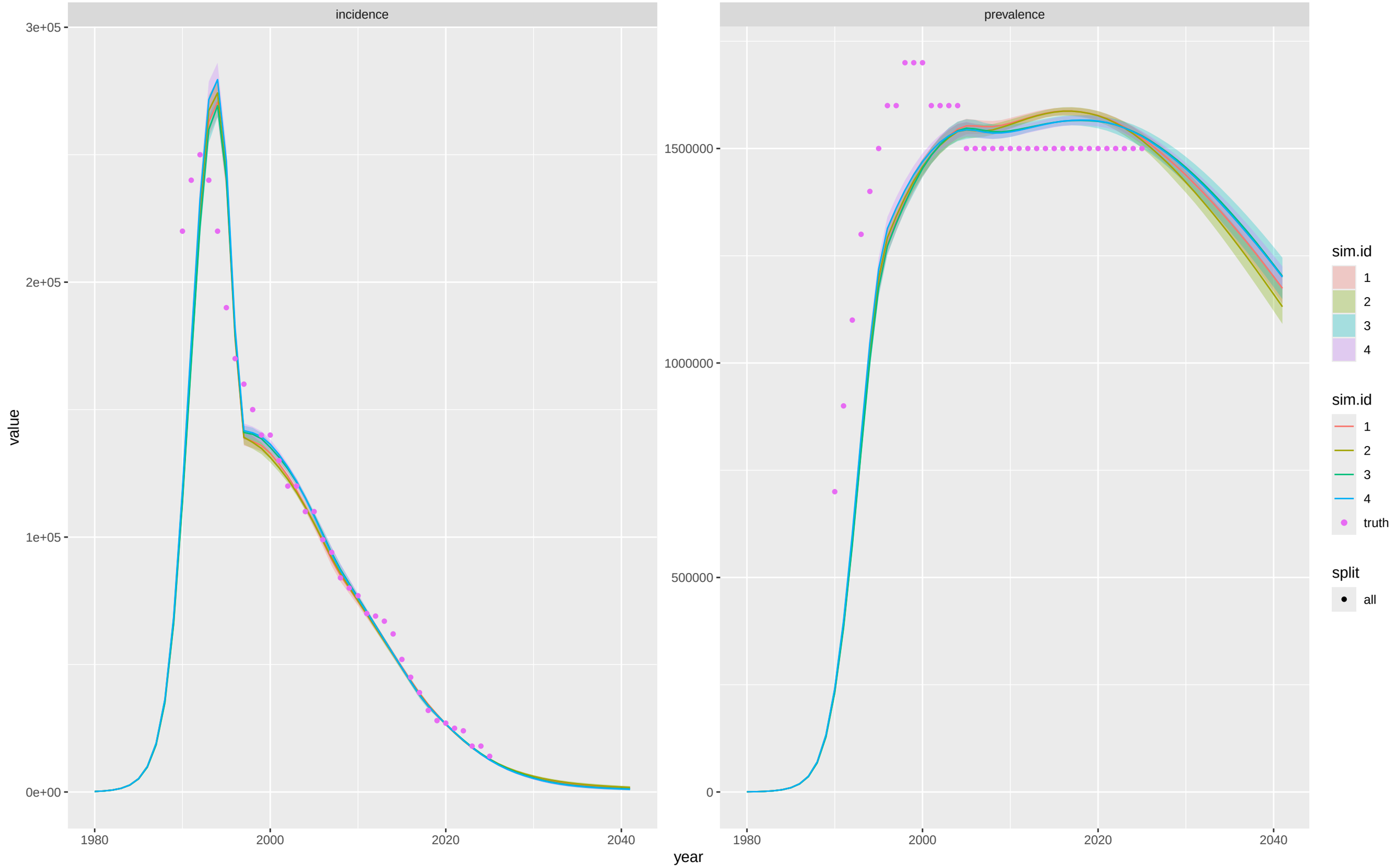
chain
1
2
3
4

type
Log Likelihood
Log Prior
Log Prior x Likelihood

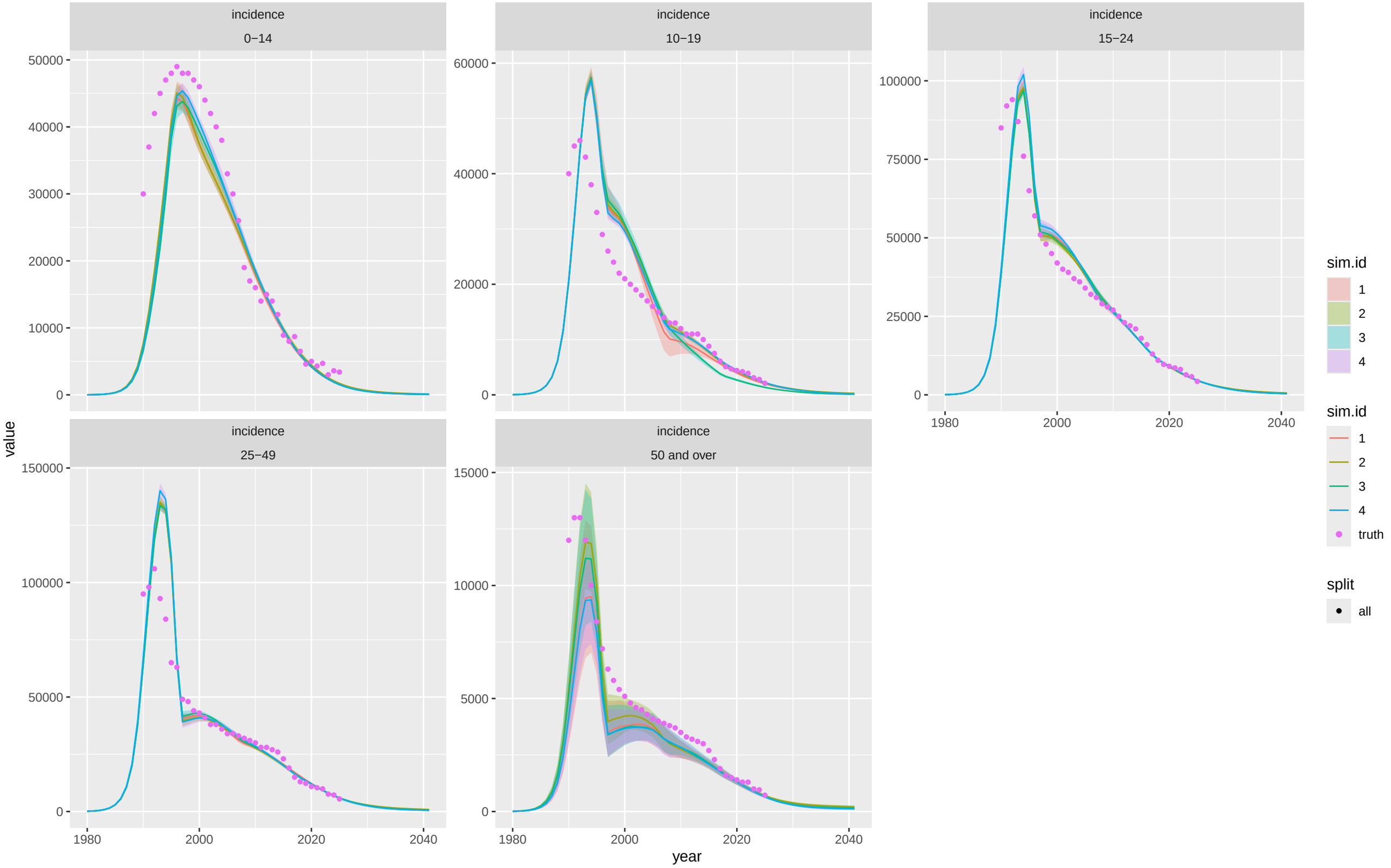
Kenya step 2: trace



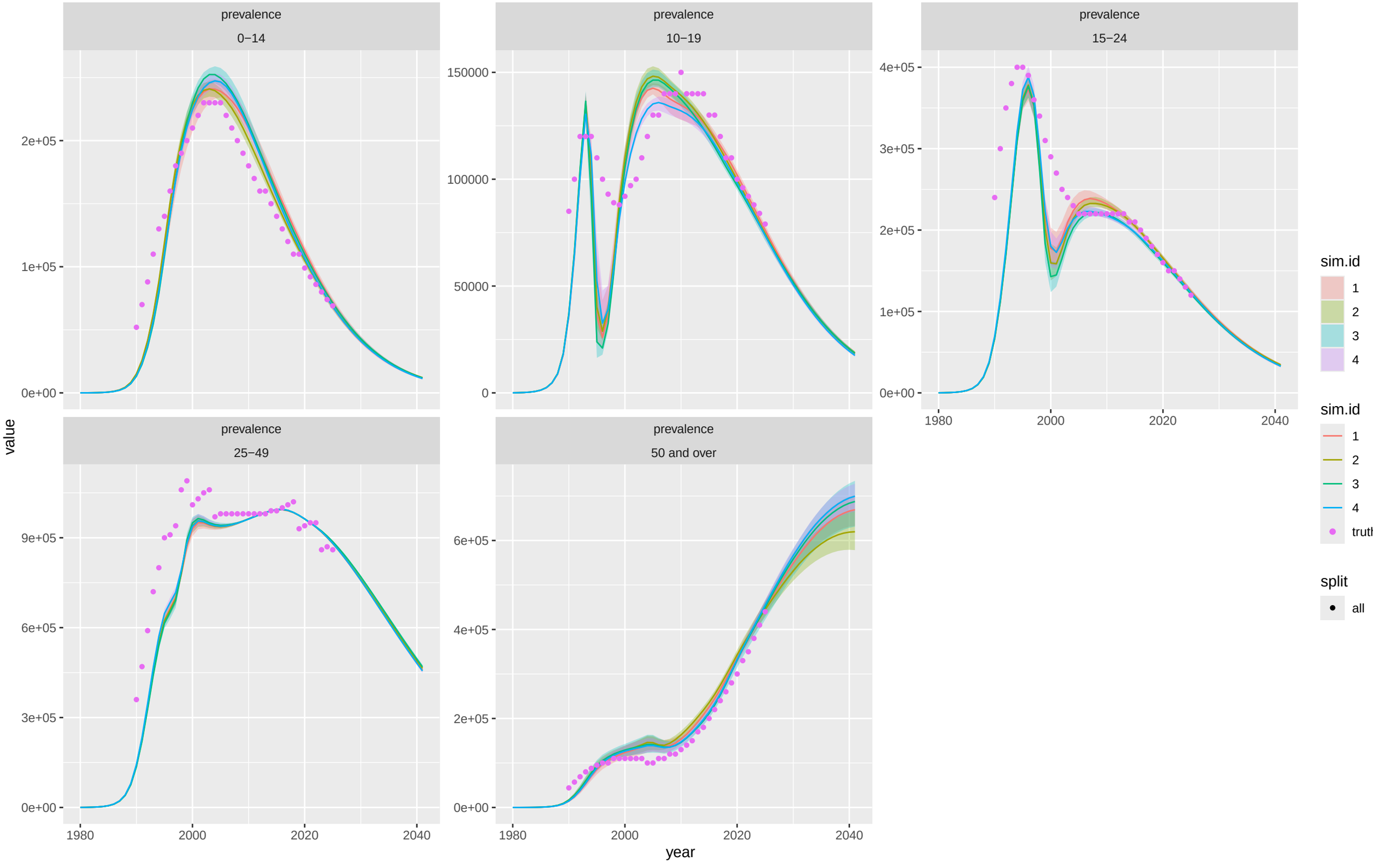
Kenya step 2: fit (one color per chain)



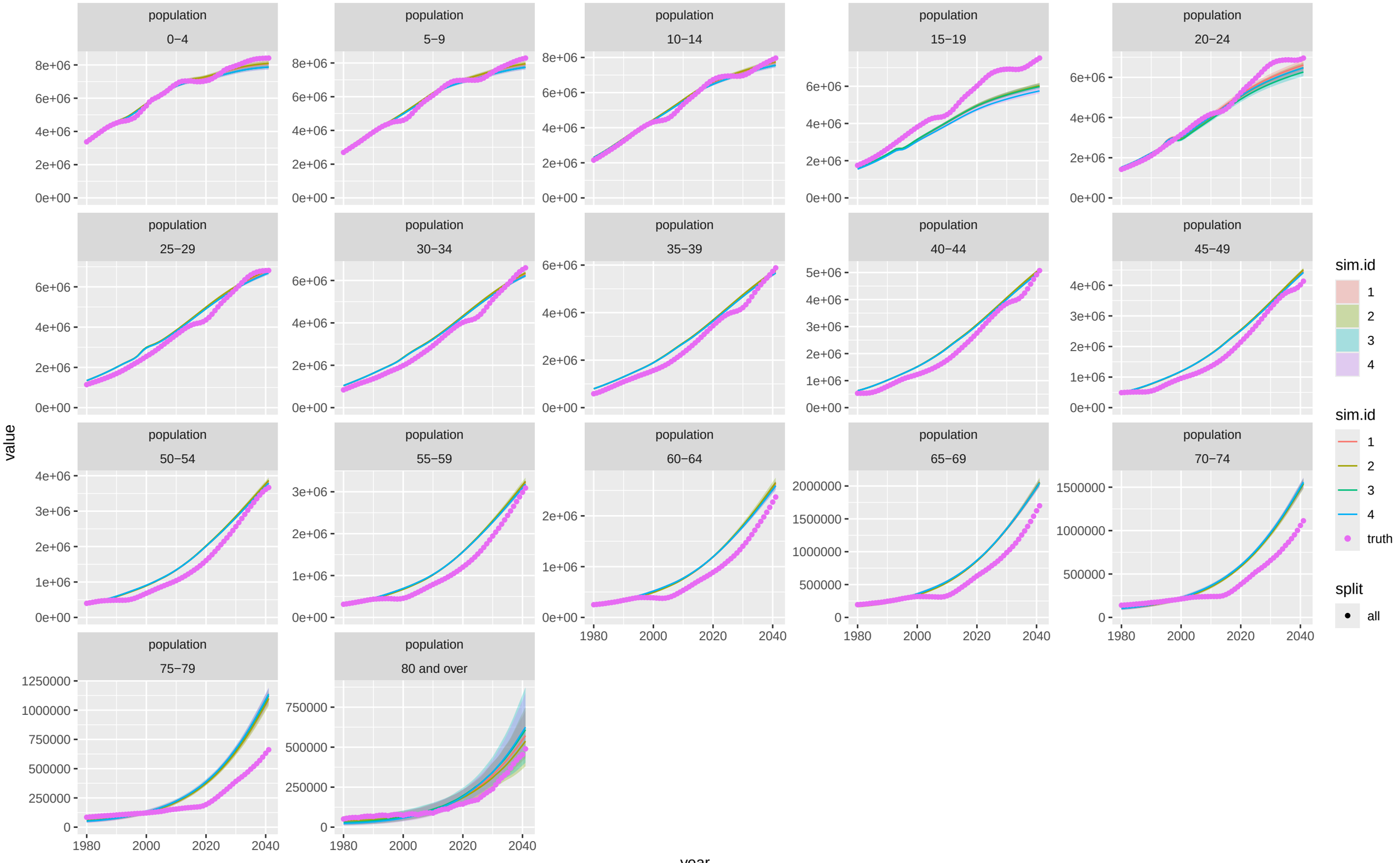
Kenya step 2: incidence.by.age (one color per chain)



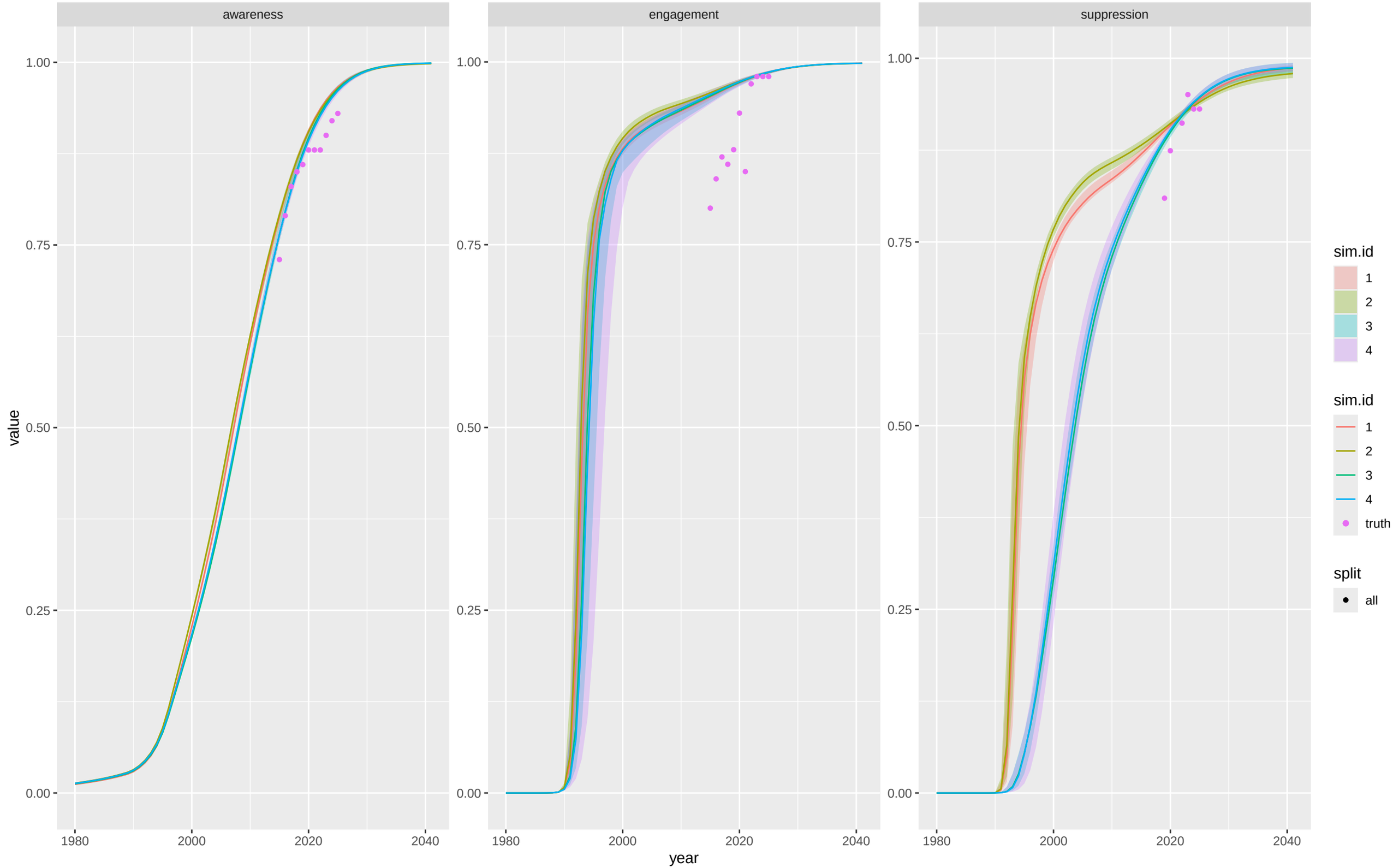
Kenya step 2: prevalence.by.age (one color per chain)



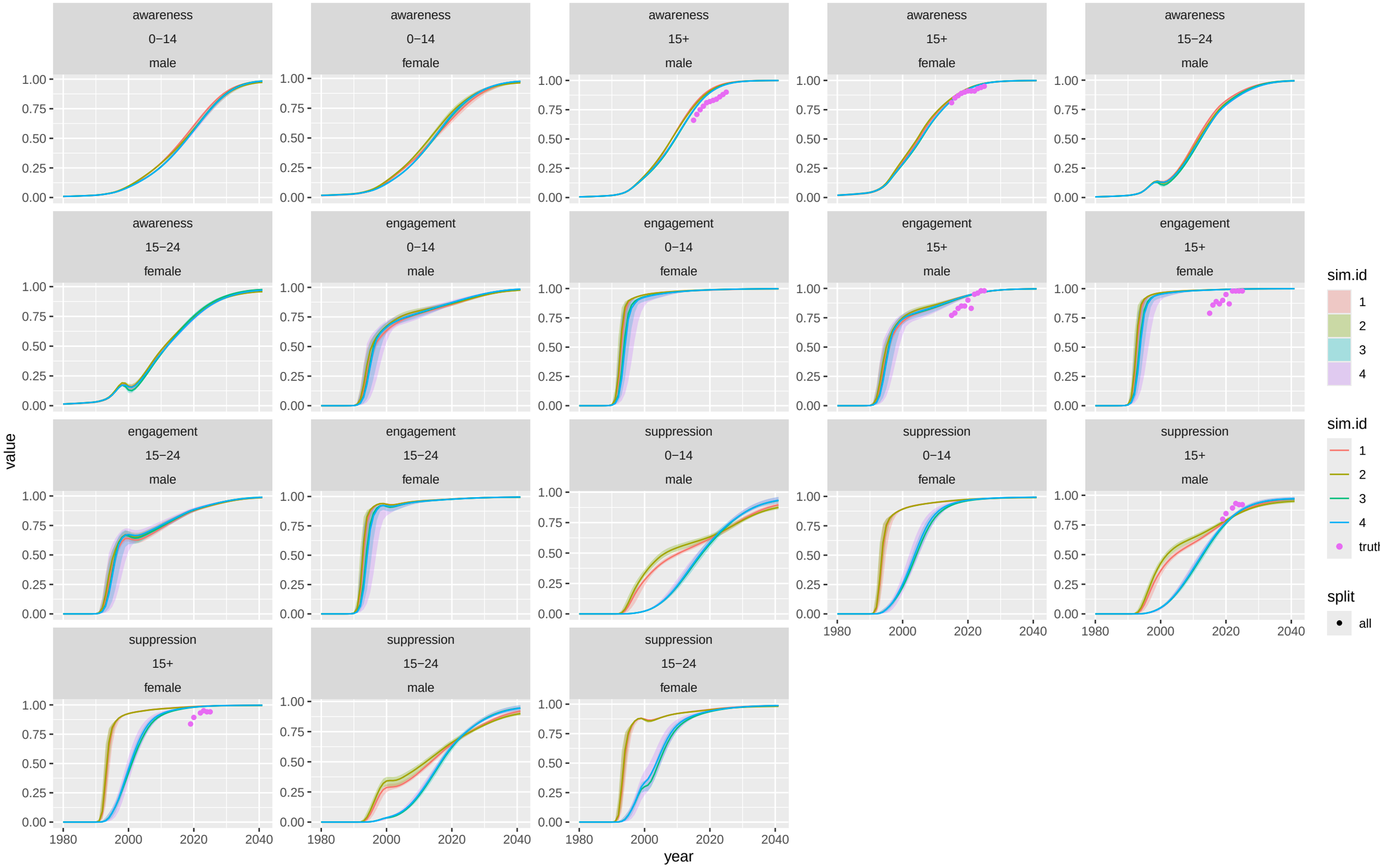
Kenya step 2: population.by.age (one color per chain)



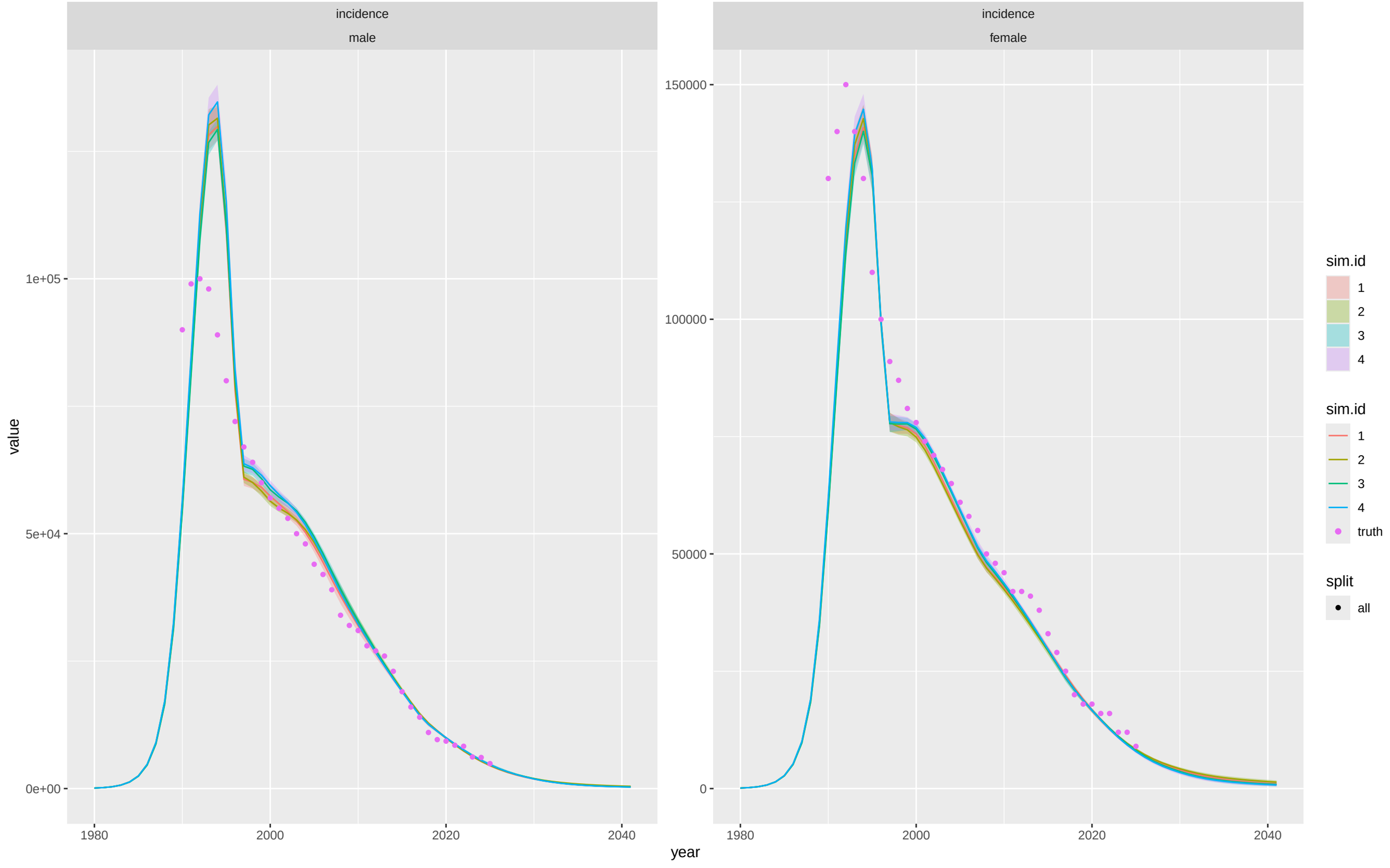
Kenya step 2: cascade (one color per chain)



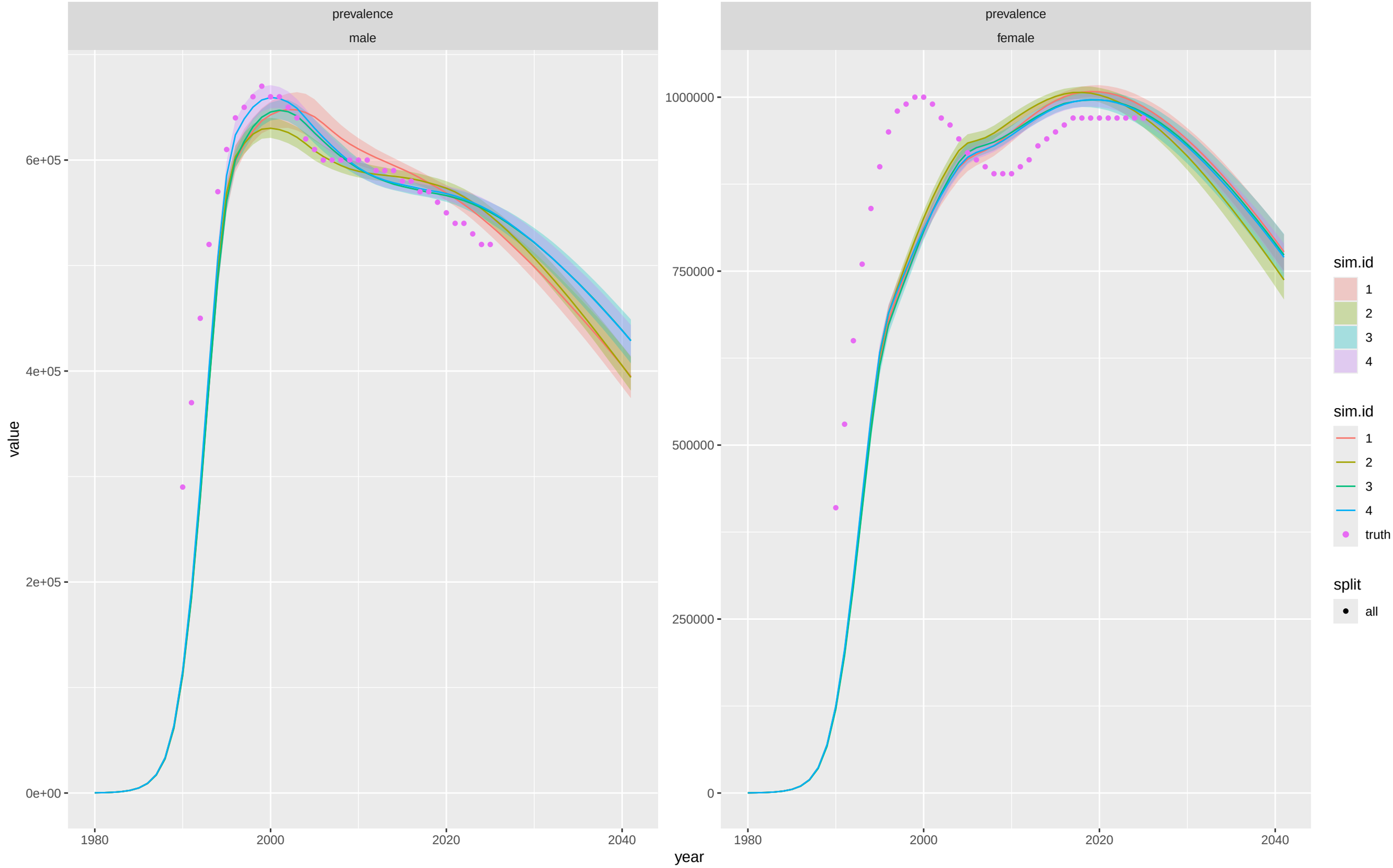
Kenya step 2: cascade.by.age.sex (one color per chain)



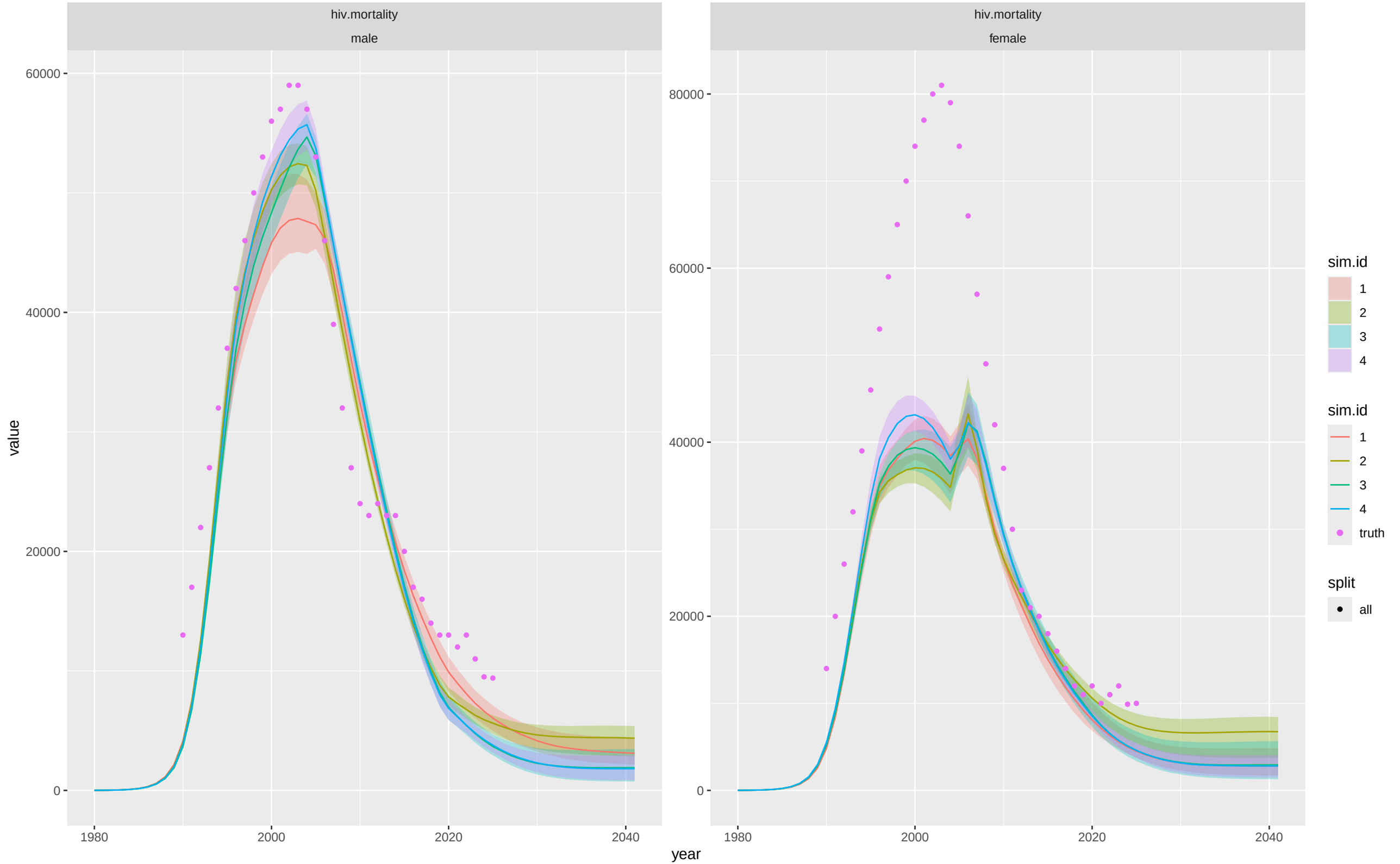
Kenya step 2: incidence.by.sex (one color per chain)



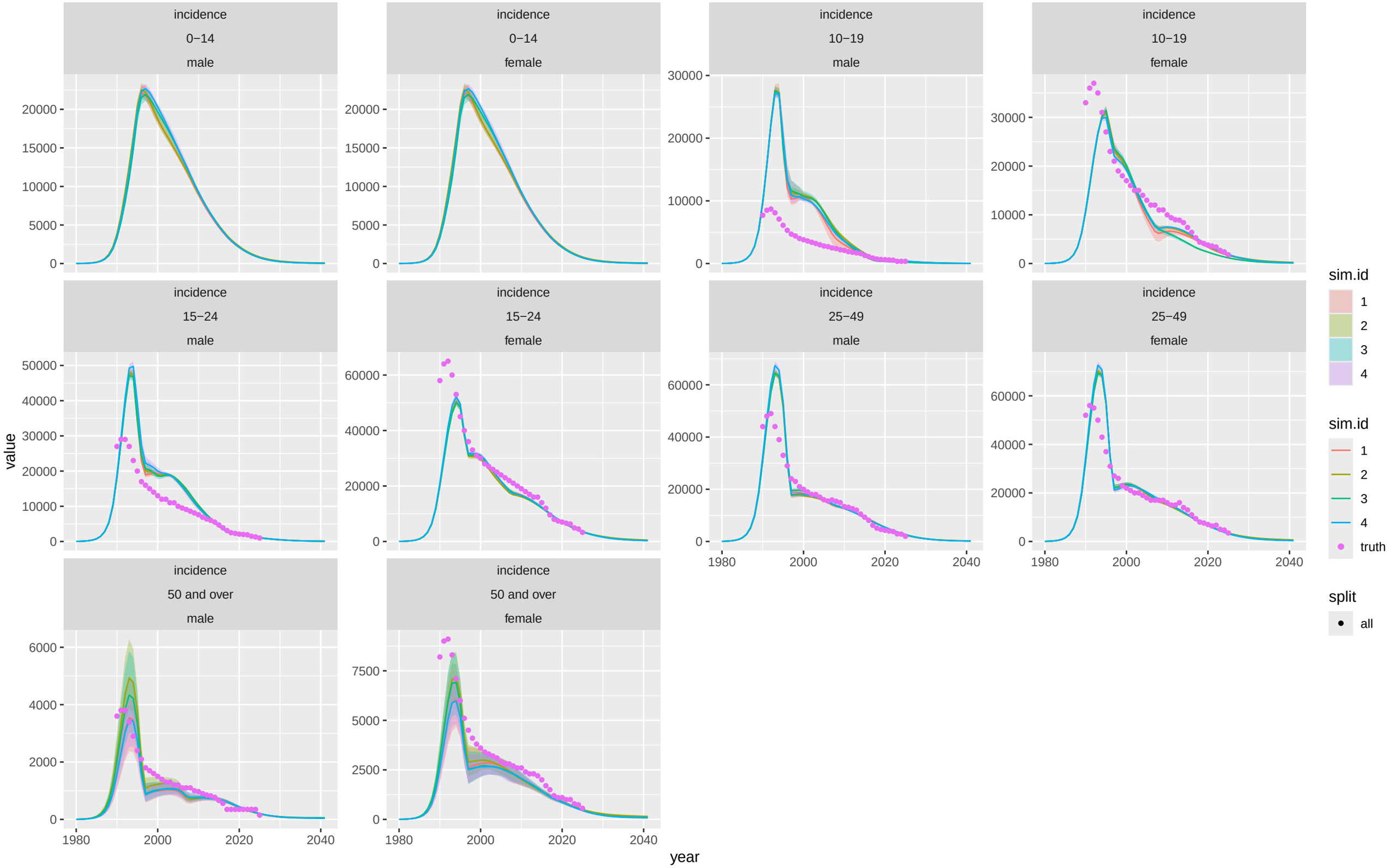
Kenya step 2: prevalence.by.sex (one color per chain)



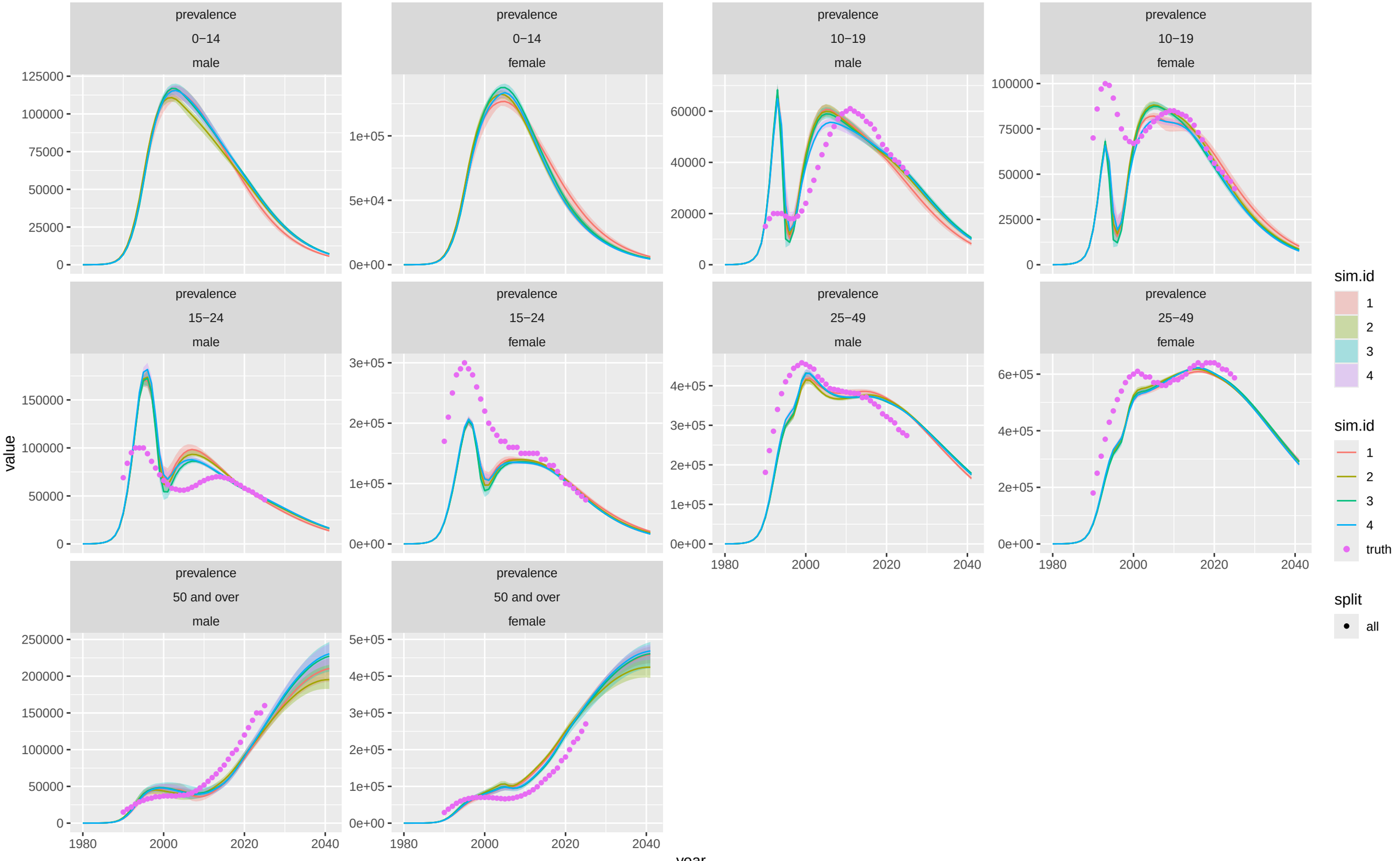
Kenya step 2: hiv.mortality.by.sex (one color per chain)



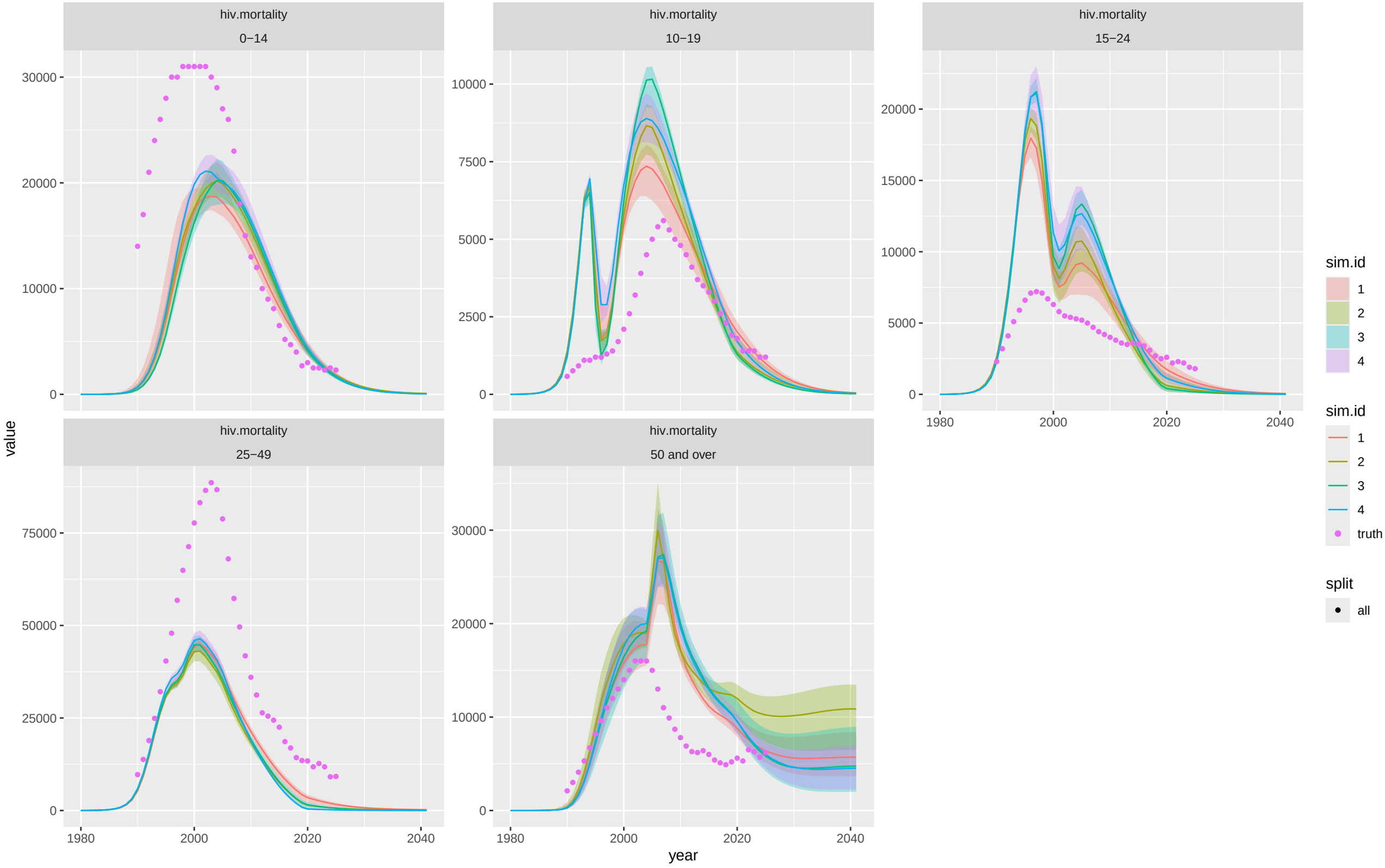
Kenya step 2: incidence.by.age.sex (one color per chain)



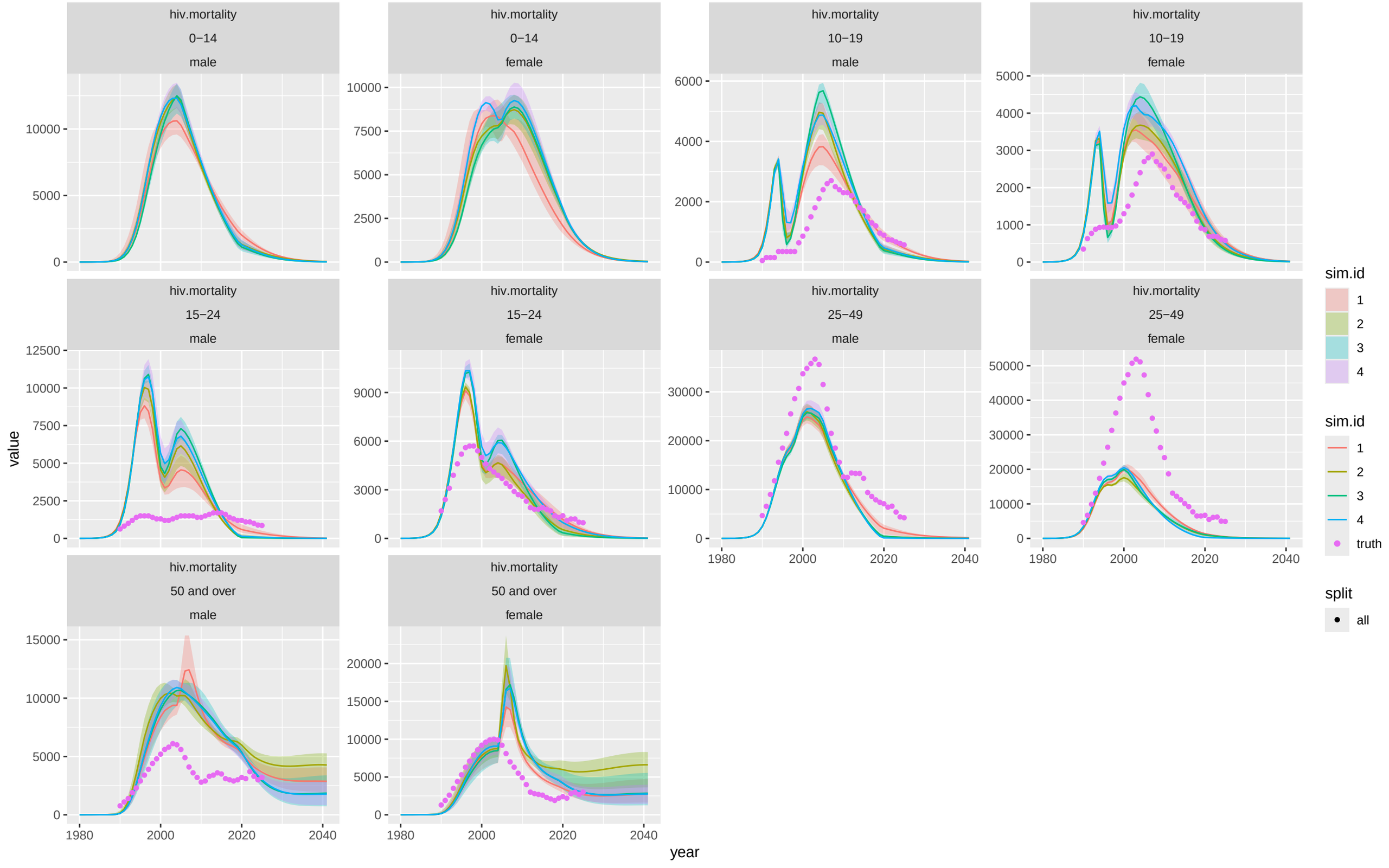
Kenya step 2: prevalence.by.age.sex (one color per chain)



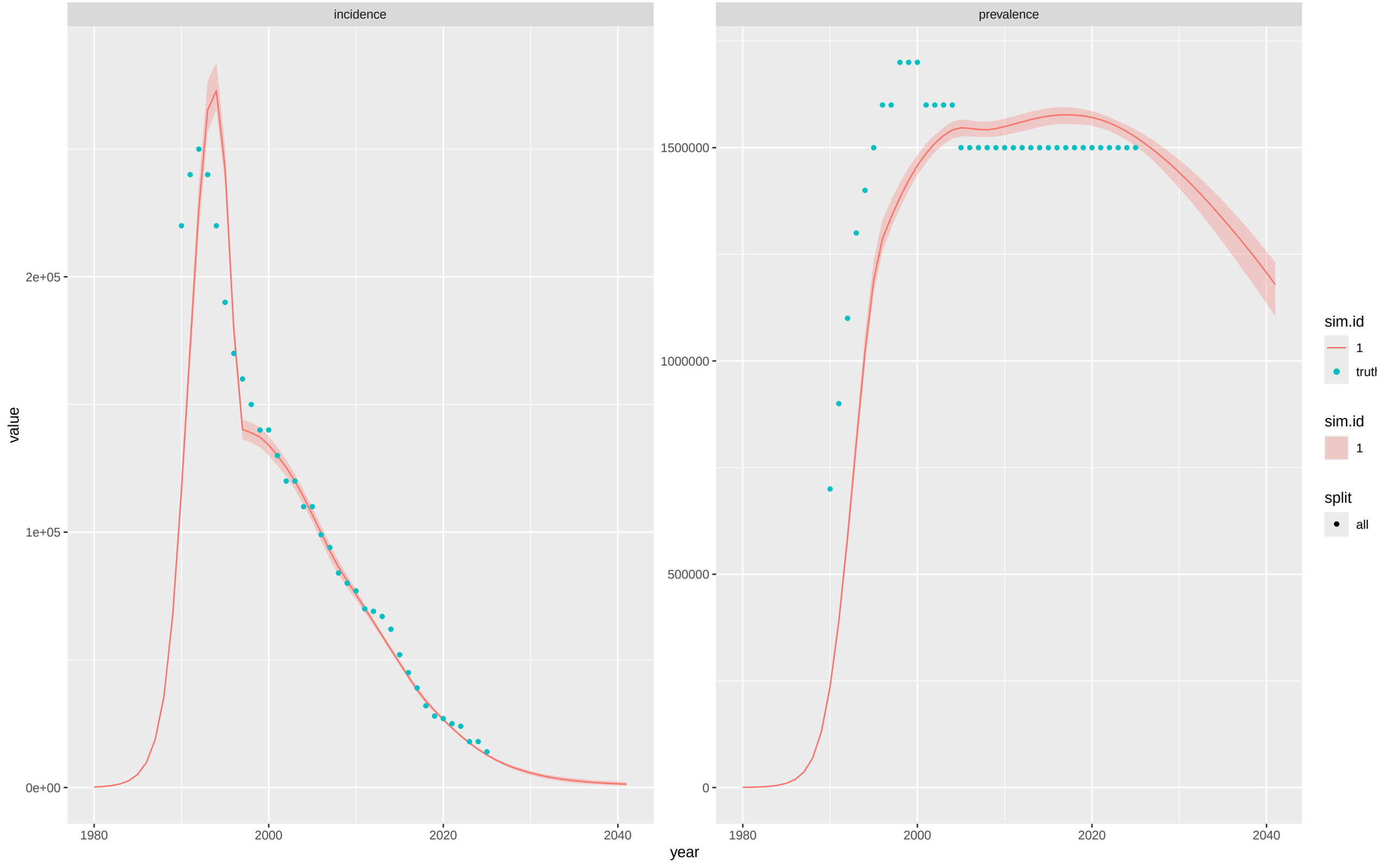
Kenya step 2: hiv.mortality.by.age (one color per chain)



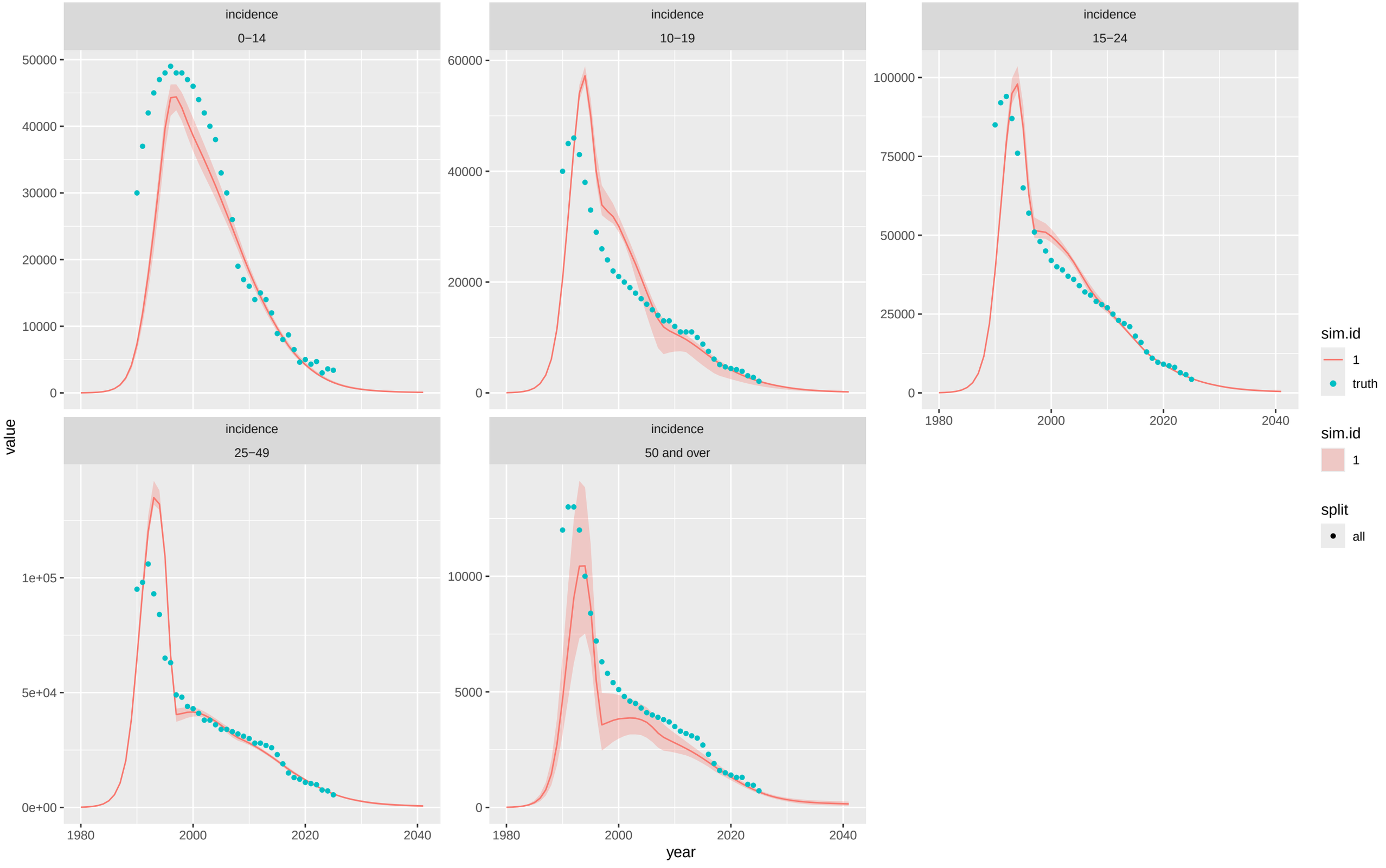
Kenya step 2: hiv.mortality.by.age.sex (one color per chain)



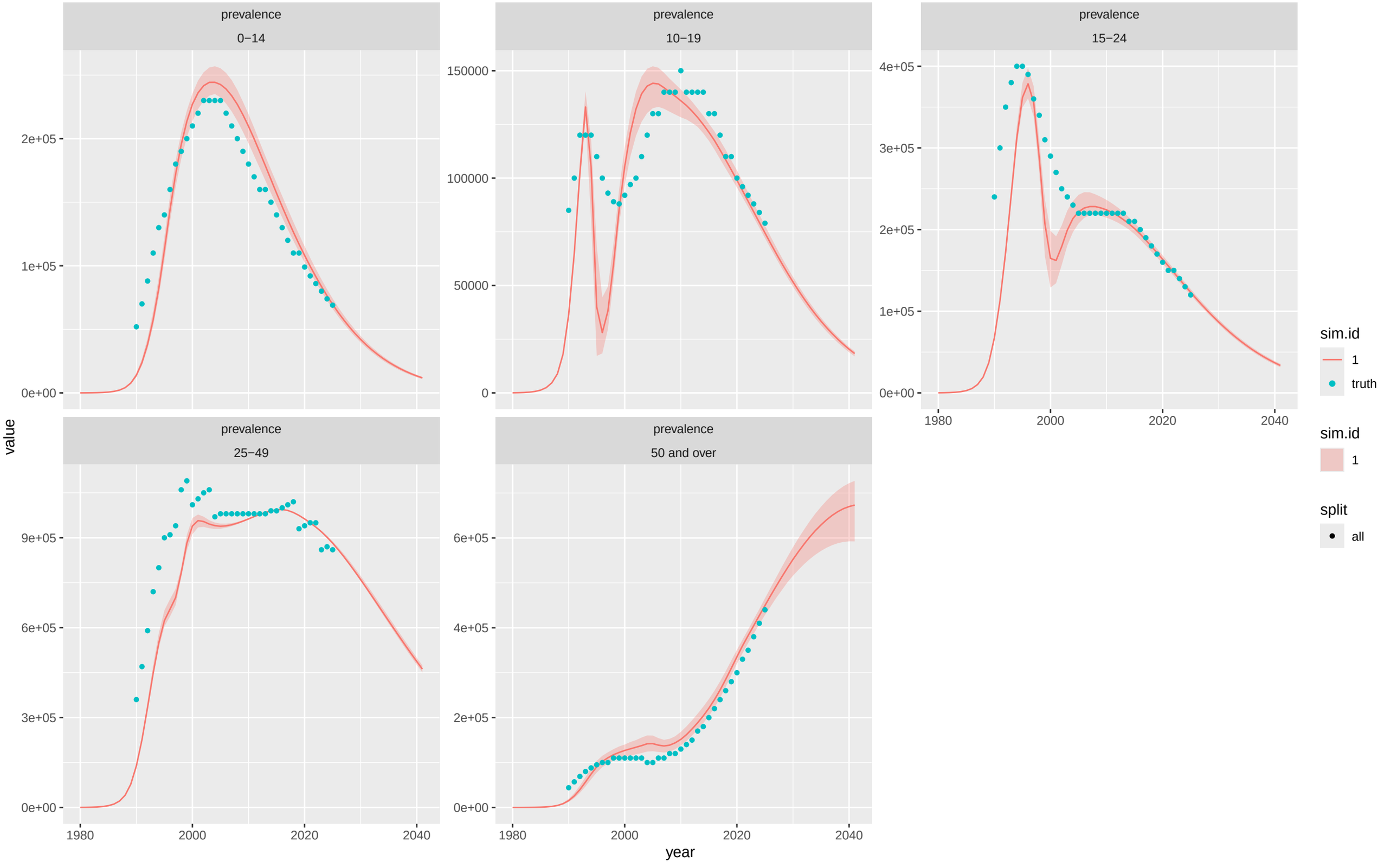
Kenya: fit (merged Step 2 simset)



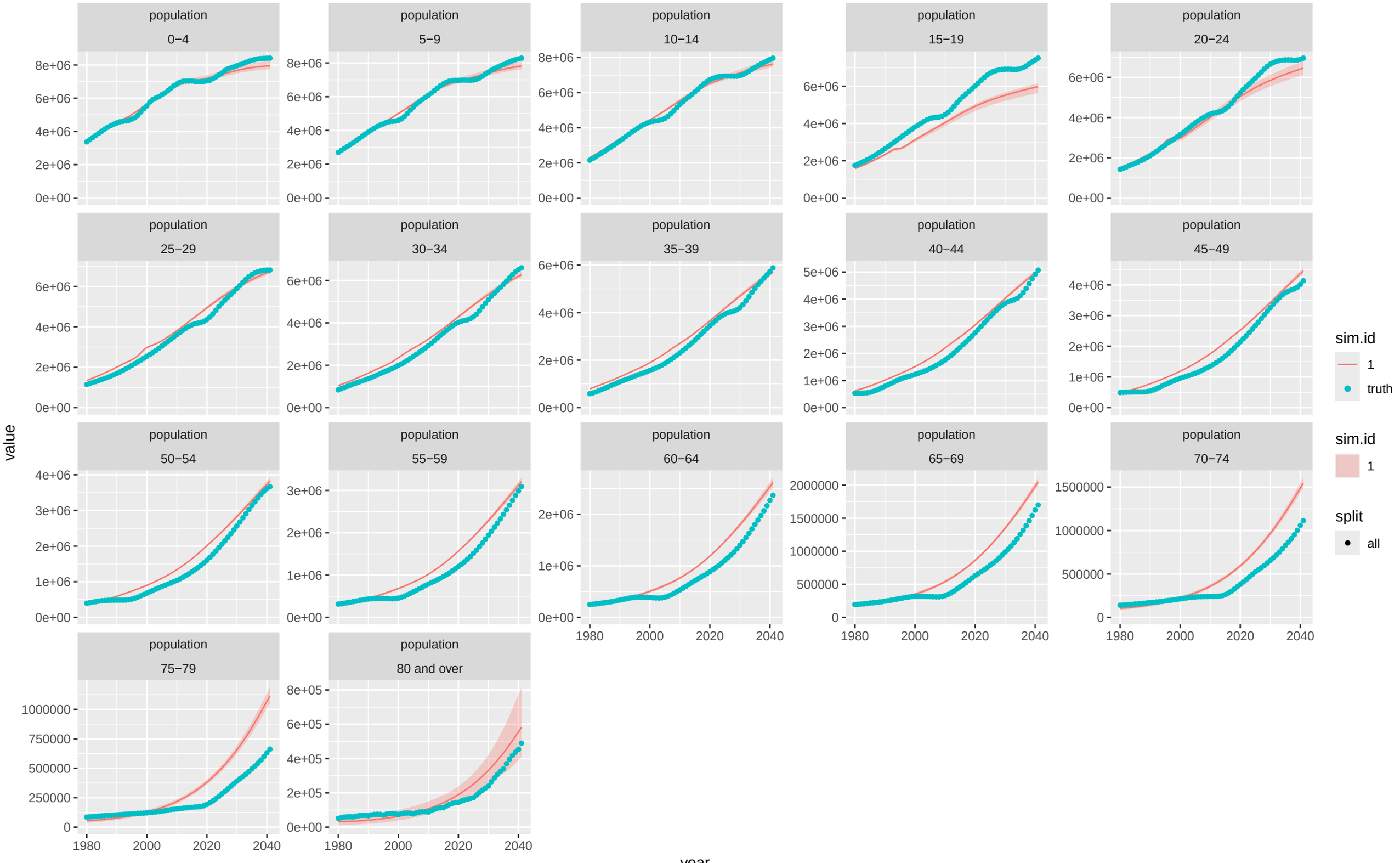
Kenya: incidence.by.age (merged Step 2 simset)



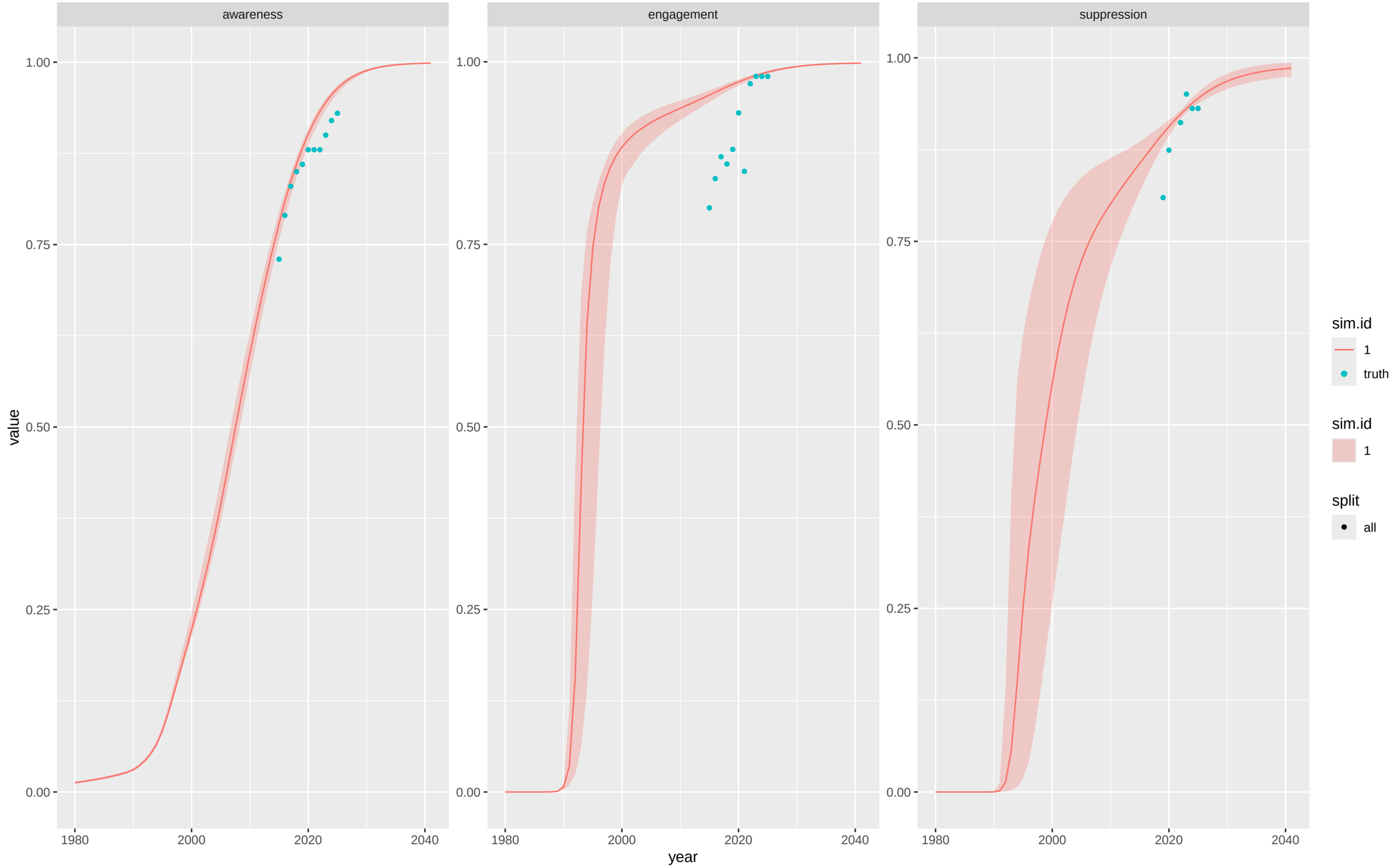
Kenya: prevalence.by.age (merged Step 2 simset)



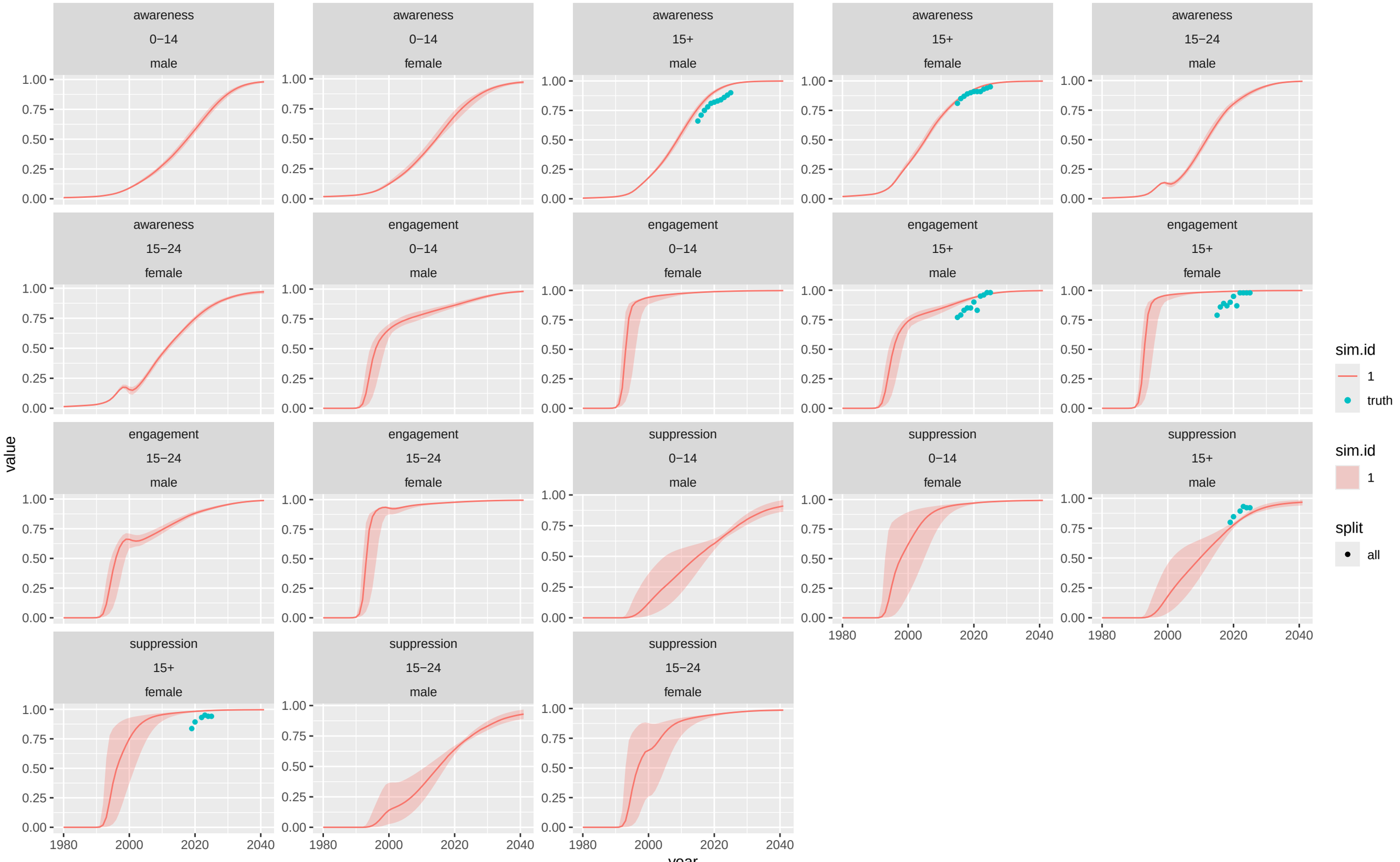
Kenya: population.by.age (merged Step 2 simset)



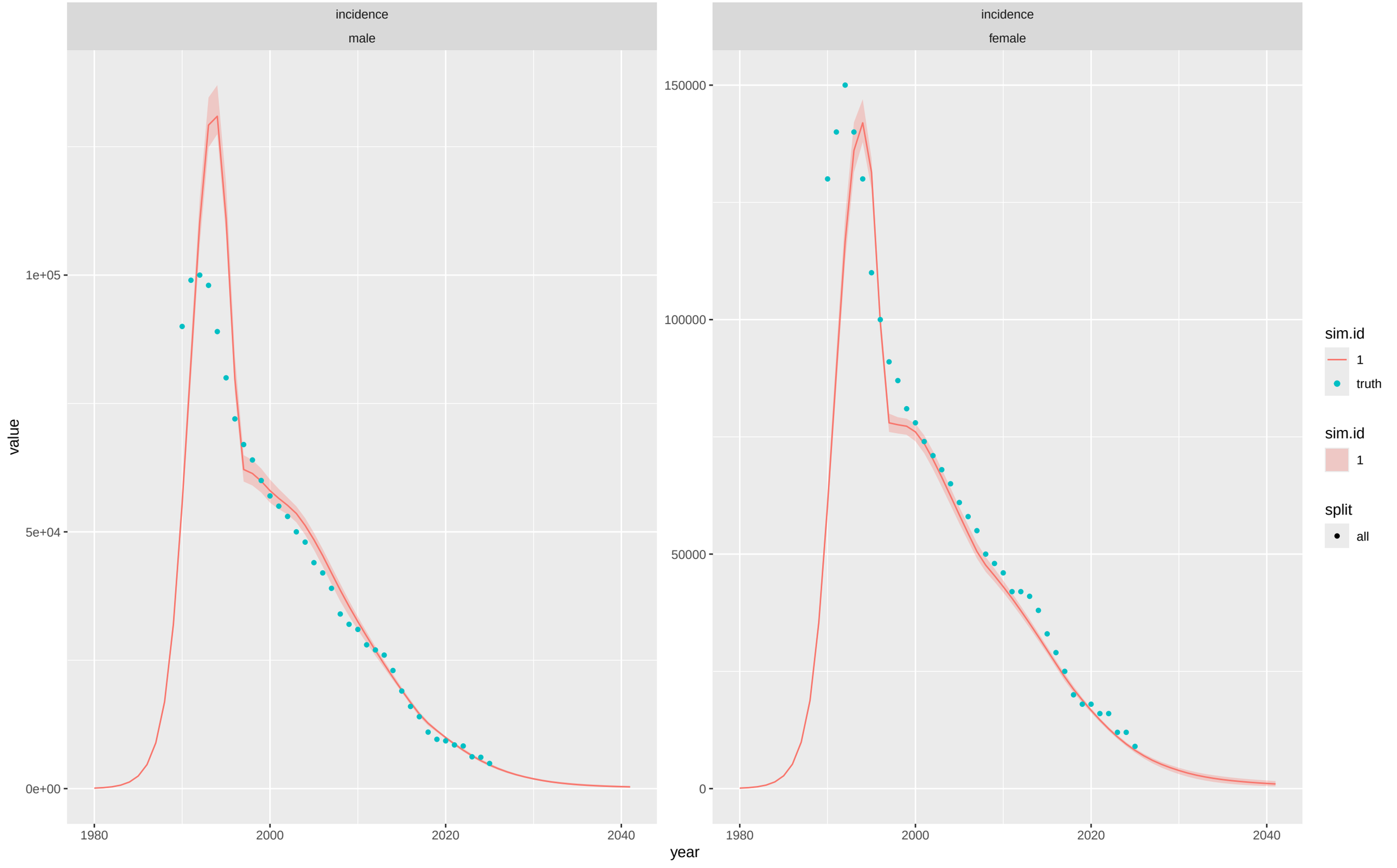
Kenya: cascade (merged Step 2 simset)



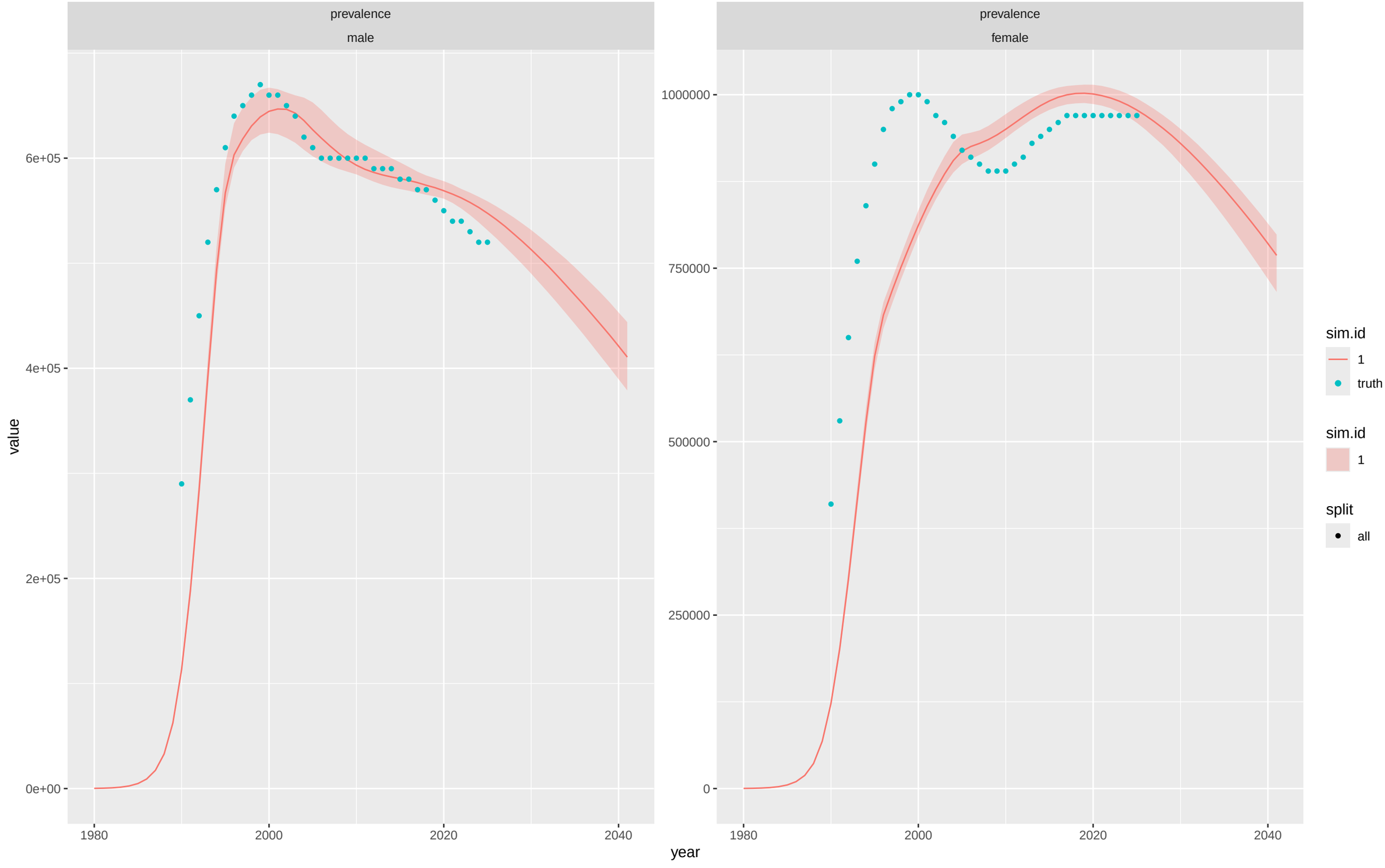
Kenya: cascade.by.age.sex (merged Step 2 simset)



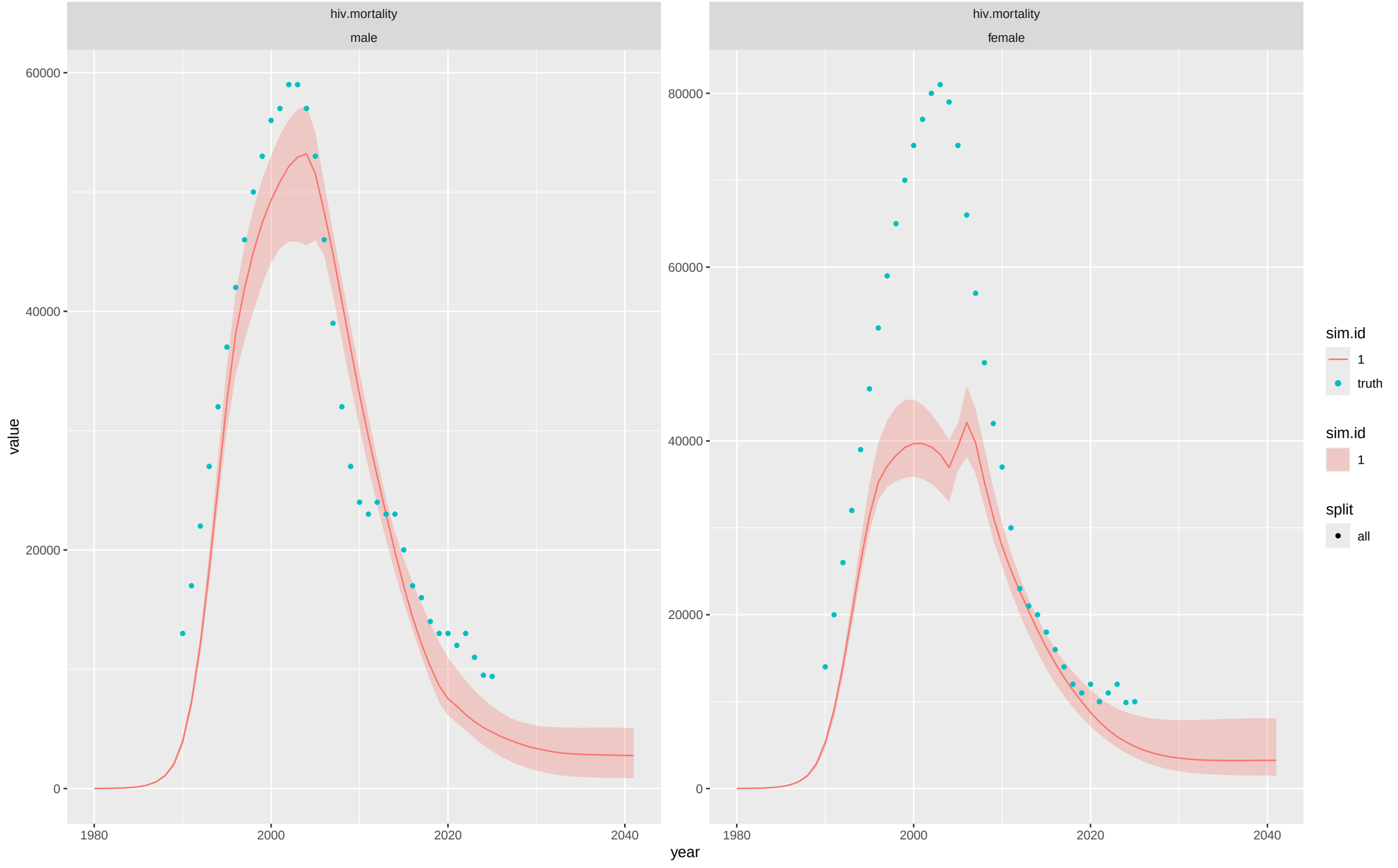
Kenya: incidence.by.sex (merged Step 2 simset)



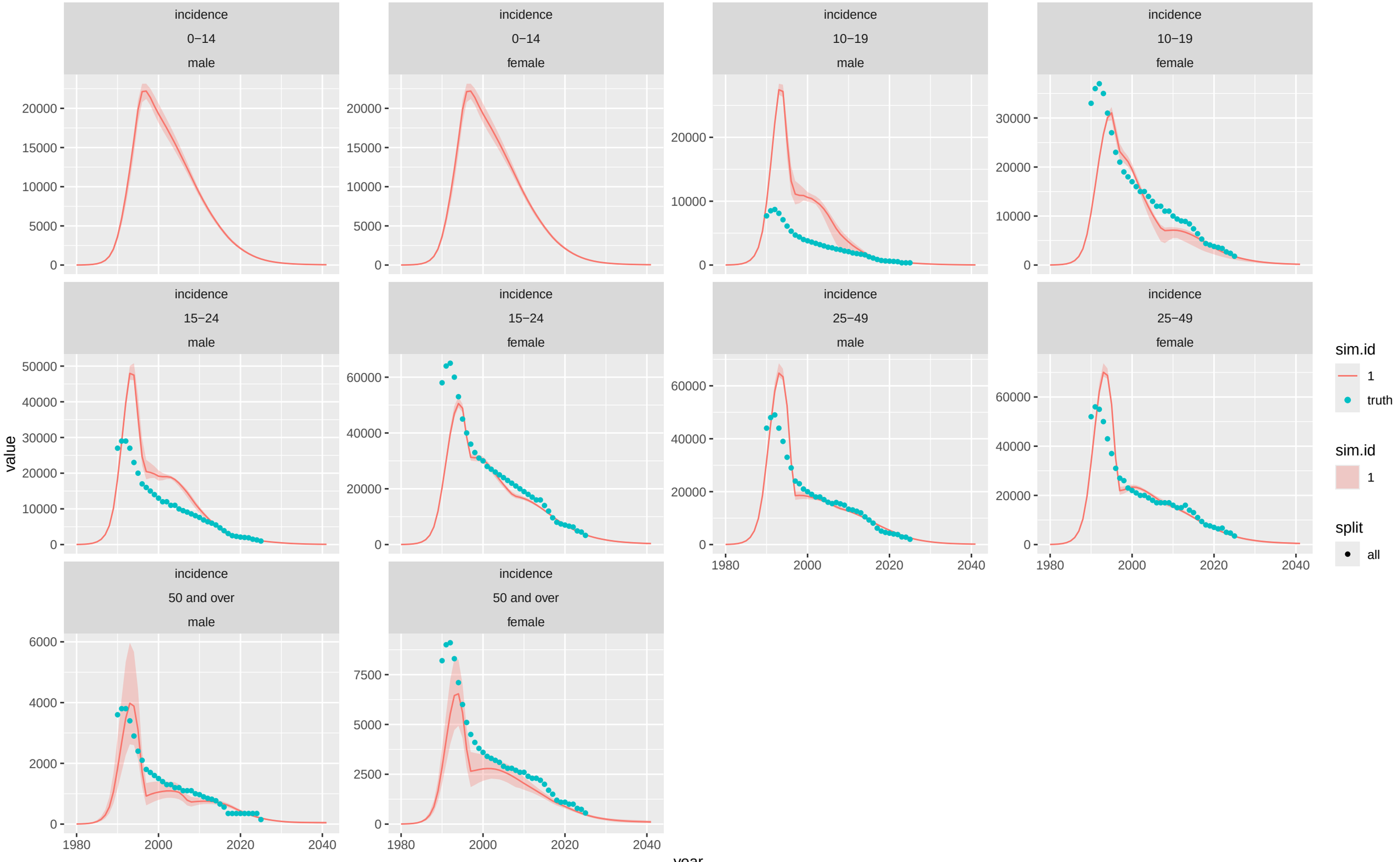
Kenya: prevalence.by.sex (merged Step 2 simset)



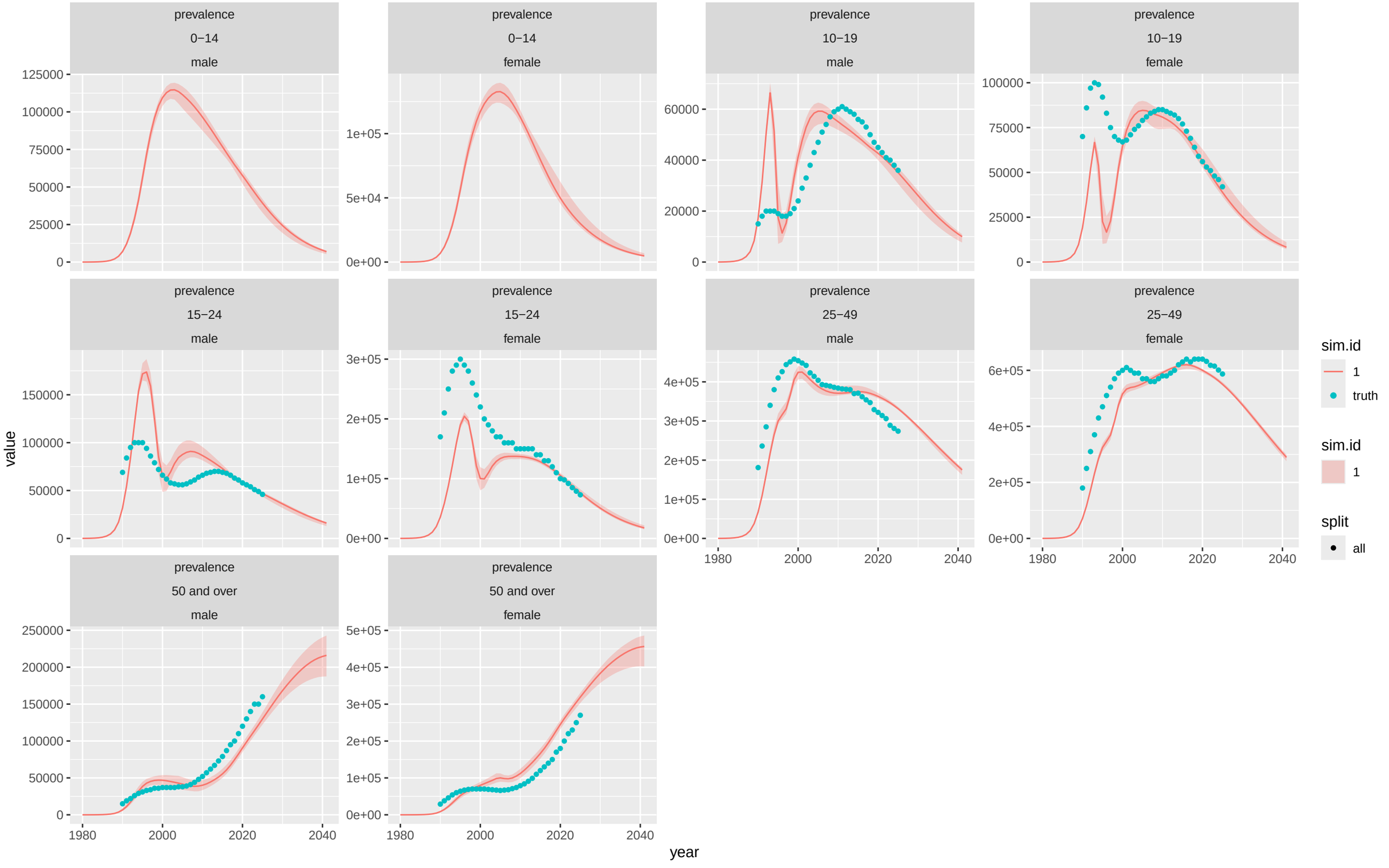
Kenya: hiv.mortality.by.sex (merged Step 2 simset)



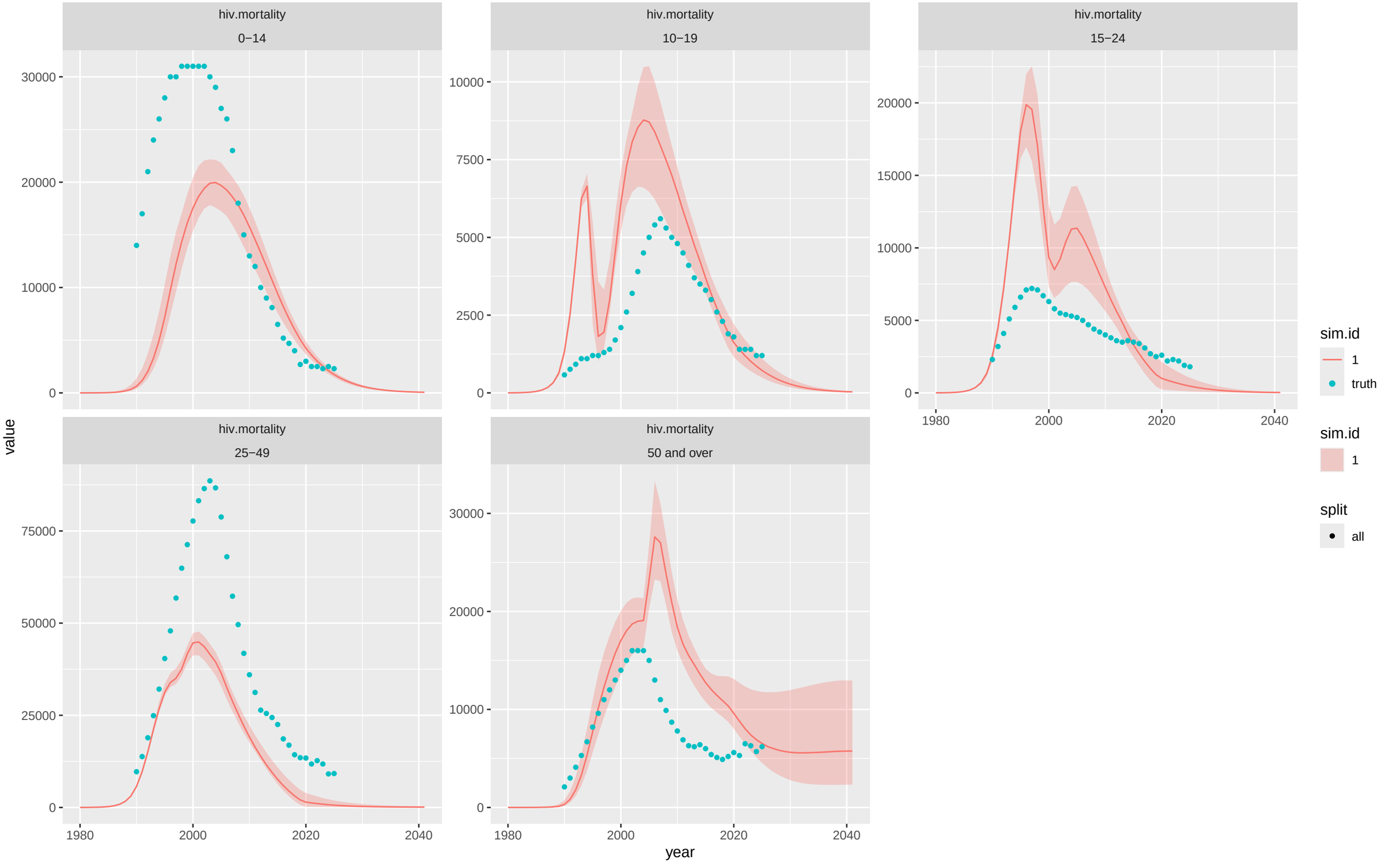
Kenya: incidence.by.age.sex (merged Step 2 simset)



Kenya: prevalence.by.age.sex (merged Step 2 simset)



Kenya: hiv.mortality.by.age (merged Step 2 simset)



Kenya: hiv.mortality.by.age.sex (merged Step 2 simset)